

Appendix

Anonymous Author(s)

NeurIPS Paper Checklist

1. For all authors...
 - (a) Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope? [Yes]
 - (b) Did you describe the limitations of your work? [Yes] Limitations and deployment constraints are discussed in the conclusion and appendix.
 - (c) Did you discuss any potential negative societal impacts of your work? [Yes] Privacy, elderly-care monitoring, and responsible deployment concerns are discussed in the appendix.
 - (d) Have you read the ethics review guidelines and ensured that your paper conforms to them? [Yes]
2. If you are including theoretical results...
 - (a) Did you state the full set of assumptions of all theoretical results? [N/A]
 - (b) Did you include complete proofs of all theoretical results? [N/A]
3. If you ran experiments...
 - (a) Did you include the code, data, and instructions needed to reproduce the main experimental results either in the supplemental material or as a URL? [Yes] Reproducibility materials are provided in the supplementary material.
 - (b) Did you specify all the training details, such as data splits, hyperparameters, and how they were chosen? [Yes] Reconstruction training, evaluation datasets, and implementation details are described in Section 4 and the appendix.
 - (c) Did you report error bars, for example with respect to the random seed after running experiments multiple times? [No] Due to computational and time constraints, experiments were not repeated across multiple random seeds.

- (d) Did you include the total amount of compute and the type of resources used, such as GPUs, internal cluster, or cloud provider? [Yes] The experimental setup and GPU resources are described in Section 4 and the appendix.
4. If you are using existing assets, such as code, data, models, or curating/releasing new assets...
- (a) If your work uses existing assets, did you cite the creators? [Yes]
 - (b) Did you mention the license of the assets? [Yes] Public datasets, benchmark sources, and prior models are cited appropriately.
 - (c) Did you include any new assets either in the supplemental material or as a URL? [Yes] The SAI-CAM implementation, spectral-packet encoding scripts, AI-VR restoration module, and semantic reasoning code are provided in anonymized supplementary material.
 - (d) Did you discuss whether and how consent was obtained from people whose data you are using/curating? [Yes] We use publicly available video datasets intended for academic research and benchmark evaluation.
 - (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? [Yes] To the best of our knowledge, the datasets do not contain identity labels, facial recognition annotations, or intentionally offensive content. The proposed framework further reduces privacy risk by avoiding persistent raw RGB video storage.
5. If you used crowdsourcing or conducted research with human subjects...
- (a) Did you include the full text of instructions given to participants and screenshots, if applicable? [N/A]
 - (b) Did you describe any potential participant risks, with links to Institutional Review Board approvals, if applicable? [N/A]
 - (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? [N/A]